

DFN Modelling and Parameter Calibration for the A123 Cell

A123 ANR26650M1B 2.5 Ah LFP graphite cell

The calibrated model completes all four charge records and the 1C discharge record. Their full-record voltage RMSE values range from 20.07 to 27.35 mV. HPPC reaches 18.2736 hours of the 25.1950-hour record before a modelled 2.00 V cutoff, covering 72.53% of the requested time. This report documents the baseline, initial 1C calibration, staged multi-profile search and the physical-state evidence explaining the remaining HPPC limitation.

1 Objective and available data

The assignment required a Doyle-Fuller-Newman model in PyBaMM for the A123 ANR26650M1B cell, selection of a suitable parameter set, simulation of the supplied current profiles, residual analysis, a parameter-optimization pipeline and assessment of the resulting parameter set. The required optimization inputs were 1C, 2C, 3C and 4C charge records and one 1C discharge record. HPPC supplied a demanding pulse-response assessment. [1]

Table 1 Supplied experimental records

Record	Raw rows	Duration	Role
Charge 1C	6062	102.35 min	Training
Charge 2C	4423	74.04 min	Training
Charge 3C	3844	64.45 min	Training
Charge 4C	3523	59.43 min	Training
Discharge 1C	393	64.68 min	Training
HPPC	55793	25.1950 h	Pulse assessment and acceptance

The brief permitted literature OCP relations, manual parameter tuning, grouping of parameters and external resources when their use was justified. Temperature effects could be ignored because the supplied data did not cover a temperature range. We therefore used an isothermal model at 296.15 K and did not estimate temperature dependence or degradation parameters. [1]

Recorded current was replayed as an input; measured voltage was used for comparison. HPPC also included a cumulative Ah counter. No electrode half-cell OCP measurements or independently measured initial electrode concentrations were supplied.

2 Model selection and parameter sources

DFN was selected because the assignment explicitly requested it. Its distributed solid and electrolyte states allow analysis of high-current concentration gradients and electrode potentials, beyond a terminal-voltage fit alone. The model represents spherical particles with one representative radius per electrode and resolves position through the electrode thickness. For constant particle diffusivity, its particle balance and surface reaction boundary condition are: [2]

$$\frac{\partial c_{s,k}}{\partial t} = \frac{D_{s,k}}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial c_{s,k}}{\partial r} \right), \quad -D_{s,k} \frac{\partial c_{s,k}}{\partial r} \Big|_{r=R_k} = \frac{j_k}{F}$$

Here c_s is solid lithium concentration, D_s is diffusivity, r is particle radius coordinate, R_k is the particle radius, j_k is local interfacial current density and F is Faraday's constant. The symmetric Butler-Volmer relation connects that reaction current to electrode overpotential:

$$j_k = 2j_{0,k} \sinh\left(\frac{F\eta_k}{2RT}\right), \quad \eta_k = \phi_{s,k} - \phi_{e,k} - U_k(c_{s,k}^{\text{surf}})$$

Prada2013 was an appropriate starting parameter set because the underlying study considered an earlier A123 ANR26650m1 LFP graphite power cell with a nominal capacity of 2.3 Ah. The target M1B cell is 2.5 Ah. This provided a closer chemistry and cell-family starting point than an NMC energy-cell parameter set, while still requiring recalibration. The published Prada model is a simplified electrochemical and aging model; here its parameter information was used within PyBaMM's DFN equations. [3]

Setting the nominal capacity entry to 2.5 Ah does not by itself rescale electrode lithium storage. Physical electrode capacity depends on area, thickness, active fraction and maximum concentration. Stock Prada geometry, particle sizes and maximum concentrations were retained; electrode balance and selected effective transport and kinetic values were subsequently adjusted.

OCP equations and their provenance

The LFP charge and discharge OCP equations were taken from the supplied Afshar paper, which explicitly gives both branches on PDF page 6. They were useful because the data contain both current directions and rest periods. We do not claim that Afshar is the only available hysteresis description. The exact adopted equations are: [4]

Afshar's experimental LFP cell was different from the target A123 cell. The transfer was justified by the same cathode chemistry and the explicit availability of both branch equations. Agreement with the supplied full-cell records provides an assessment of that approximation, rather than independent verification of the target cathode OCP.

$$U_{p, \text{ch}}(y) = 3.4510 - 0.009y + 0.6687e^{-35y} - 0.5e^{-210(1-y)}$$

$$U_{p, \text{dis}}(y) = 3.4077 - 0.020269y + 0.5e^{-200y} - 0.9e^{-30(1-y)}$$

The graphite equation was the Chen2020 relation already reused by PyBaMM's Prada2013 implementation. Its exponential and hyperbolic-tangent terms retain the plateau and staging structure needed to represent graphite potential. Chen characterized an LG M50 graphite-SiOx negative electrode, so the curve is not an independently verified A123 graphite OCP. It was retained as a literature approximation rather than treated as a measured property of the target cell. [5,6]

$$U_n(x) = 1.9793e^{-39.3631x} + 0.2482 - 0.0909\tanh[29.8538(x - 0.1234)] \\ - 0.04478\tanh[14.9159(x - 0.2769)] - 0.0205\tanh[30.4444(x - 0.6103)]$$

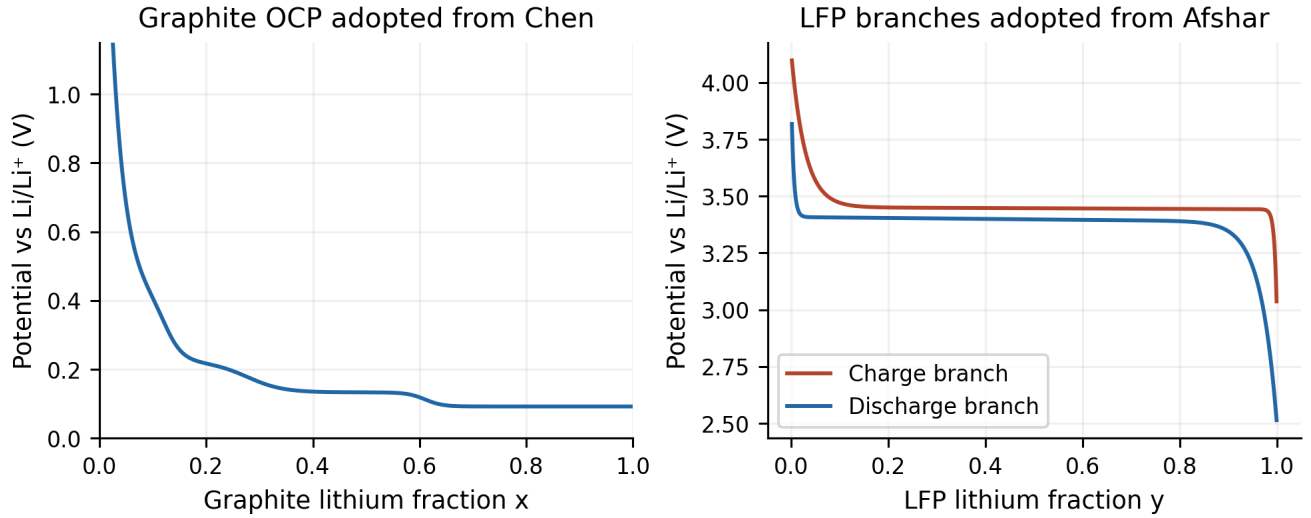


Figure 1 Adopted electrode OCP relations. Potentials are referenced to Li/Li⁺. Graphite uses Chen2020; LFP uses the separate Afshar charge and discharge branches. The graphite axis focuses on potentials relevant to the operating range.

A separate controlled OCP screen was conducted with the pre-1C parameter set, 10 mΩ resistance and voltage-based initialization. Ecker2015, Dualfoil/Marquis2019, Mohtat2020, Ramadass2004 and Safari/Schimpe expressions were tested, together with three diagnostic broadenings of Chen’s staging terms. Chen gave the lowest full-record 1C charge RMSE among these nine fixed-parameter candidates: 111.29 mV. Dualfoil extended HPPC to about 18.27 h but increased 1C charge RMSE to 214.69 mV. Some other curves lowered the reached HPPC error, but none gave a consistent improvement across the required records. These are screening results under that distinct pre-fit setup, not the calibrated errors reported later.

Without target-cell half-cell measurements, replacing a recognizable staging relation simply to smooth a full-cell charge trace would leave its physical accuracy uncertain. The adopted graphite equation is a compromise among the tested fits. Its low-lithiation shape remains relevant to the final HPPC limitation.

Exchange current density and baseline assumptions

The PyBaMM Prada2013 implementation is a combination of sources. It uses Chen2020 for the graphite exchange-current function and Kashkooli2017 for the LFP exchange-current function. Both are concentration-dependent functions. The baseline instead uses the constants published in Prada’s Table IV: $j_{0,n} = 7.5 \text{ A/m}^2$ and $j_{0,p} = 0.05 \text{ A/m}^2$. This restores the kinetic assumptions of the selected starting study. The justification is source consistency and cell-specific calibration; the presence of an LG M50 function alone does not prove that the target pulse response will fail. [3,6]

The model includes an added effective resistance term, electrolyte transport, reaction kinetics and particle diffusion. The baseline added resistance is 10 mΩ. Subsequent fitting enforces an 8 mΩ floor as specified for this study. This floor applies to the added resistance parameter; total cell polarization also contains the DFN’s internal losses.

The published Prada exchange-current constants are treated as effective constants, rather than independently remeasured target-cell kinetics. Constant diffusivities, fixed representative particle sizes and uniform initial concentrations are additional modelling assumptions. Temperature evolution, aging and a particle-size distribution were not included.

3 Data processing and initial states

Time was shifted to zero for each record, duplicate timestamps were removed and the remaining time axis was checked for increasing order. There was one duplicate in the discharge file and 333 in HPPC; the four charge files had no duplicate timestamps. HPPC retained 55460 unique measurement times. The assessment scripts retained the last duplicate; the reproduced search retains its original training-file loading convention. The resulting assessment differences are at most 0.00765 mV in RMSE. CSV current is positive on charge, whereas PyBaMM current is positive on discharge, so the sign was reversed for simulation.

The four charge recordings have gaps of approximately one second or less. Discharge is much more sparsely sampled, with gaps up to 10.019 s. HPPC has 13 intervals longer than 30 s; the longest is 267.340 s. Ordinary linear interpolation between the endpoint currents can misrepresent charge passed during these intervals and shift the simulated electrode states.

For long HPPC gaps, the cumulative Ah difference constrains the average current: $I_{\text{average}} = 3600 \Delta\text{Ah} / \Delta t$. A plateau with short transition ramps was inserted and its integral matched to that charge difference. For informative sparse switching intervals, a two-level current with an inferred switching time was used. The audit records 13 long-gap repairs and 180 sparse-transition repairs. Implausible charge/current combinations are rejected rather than silently accepted.

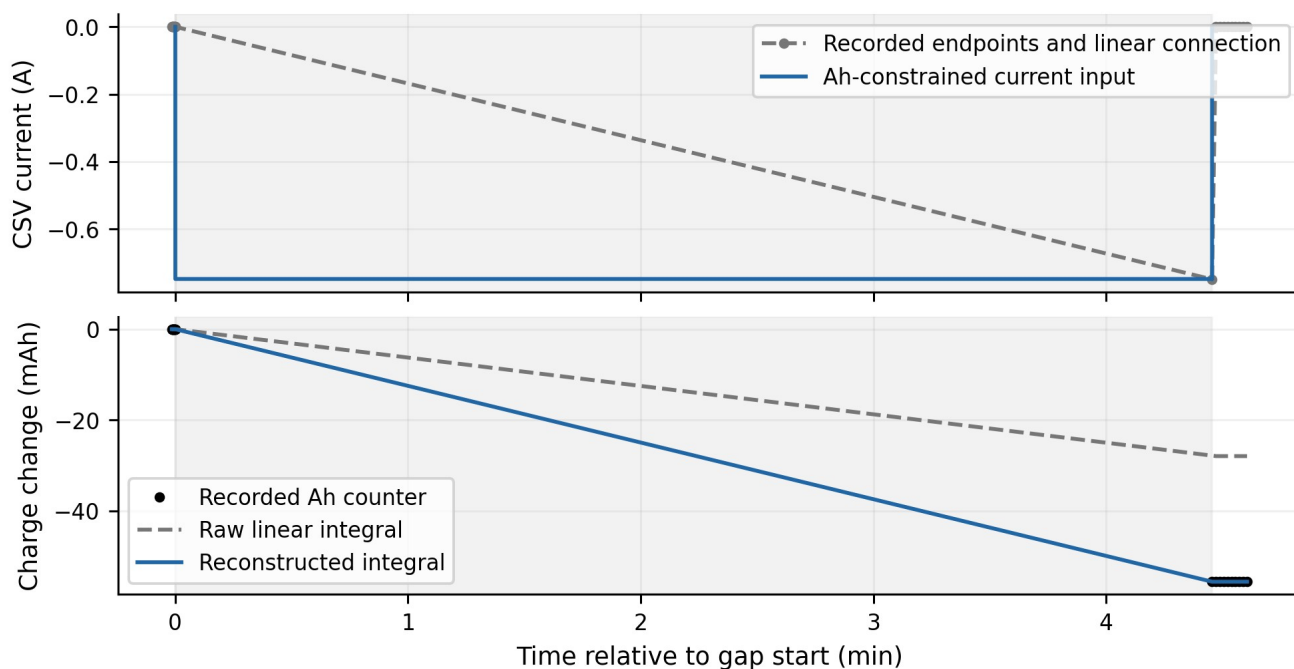


Figure 2 An example of the longest HPPC gap. The current input is reconstructed using the Ah increment. The shaded interval contains no added voltage measurements; endpoint information constrains charge but does not uniquely recover the true current waveform.

At the end of the HPPC record, the recorded counter gives 2.4375 Ah of net discharge and the reconstructed input gives 2.4323 Ah. Their difference is approximately 5.2 mAh, or 0.21%. This supports the cumulative charge consistency of the reconstruction, while local current timing and voltage within unrecorded intervals remain uncertain. Reconstructed current, charge-balance traces and the repair audit are included with the code.

Initialization using the measured rest voltage

Using the voltage at the end of the preceding 1C charge as the HPPC or discharge initial state caused inconsistent starts and early voltage events. Each record was instead initialized from its own initial rest voltage. The last ten available initial-rest voltage samples, or all samples if fewer were available, supplied the median voltage. The rest threshold was $|I| \leq 0.05$ A. HPPC's initial rest lasted only about 8.39 s, so its voltage is an approximate state constraint rather than proof of full electrochemical equilibrium.

The two electrode concentrations must also satisfy a common lithium inventory. Their maximum electrode storage capacities and initial balance are:

$$Q_k = \frac{FAL_k\epsilon_{s,k}C_{s,k}^{\max}}{3600}, \quad Q_n x_0 + Q_p y_0 = Q_{\text{Li}}$$

Here A is electrode area, L is thickness, ϵ_s is active-material fraction, and Q_{Li} is an Ah-equivalent lithium inventory. Q_{Li} is not the rated or directly measured usable cell capacity. Initial fractions x_0 and y_0 are found by a bounded root solve combining that inventory with the rest-voltage condition:

$$U_p(y_0, h_0) - U_n(x_0) = V_{\text{rest}}, \quad 0 < x_0, y_0 < 1$$

This calculation is repeated for every optimization candidate, because active fractions and inventory can change. Charge and HPPC retain an assumed charged initial LFP history. The independent discharge record has a fitted initial history fraction. Matching voltage does not establish that the assumed inventory, history or uniform concentration profiles are uniquely correct.

4 Baseline results and initial 1C calibration

The baseline uses stock Prada material and geometry information, the literature OCP equations, the restored constant exchange-current densities, 10 mΩ resistance and a graphite diffusivity of 3×10^{-15} m²/s. Its stock initial lithium inventory is 2.36705 Ah-equivalent. Before gradual hysteresis was introduced, LFP branch selection followed the current direction and retained the selected branch during rest; charge records stayed on the charge branch. This global direction-memory rule is a simpler approximation than the locally evolving history model used later.

Table 2 Baseline results on the reached portion of each record

Profile	Reached RMSE mV	Time coverage	Stopping event
Charge 1C	40.78	48.82%	OV
Charge 2C	41.03	33.31%	OV
Charge 3C	54.19	22.28%	OV
Charge 4C	64.10	15.99%	OV
Discharge 1C	219.63	65.04%	UV
HPPC	69.30	10.47%	OV

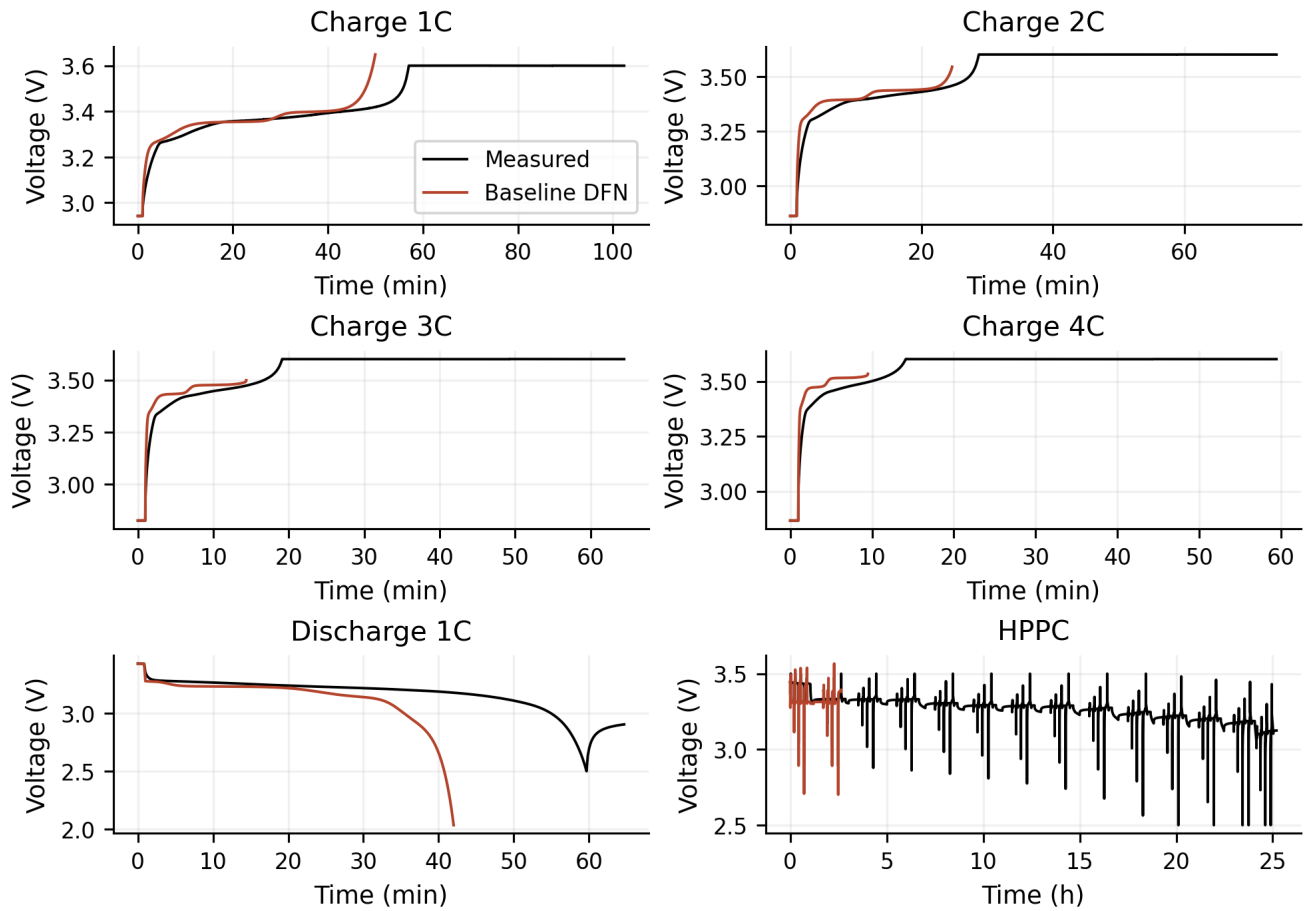


Figure 3 Baseline voltage predictions against all six experimental records. Prediction lines end at the model's voltage event. The charge records reach OV early, discharge reaches UV early, and HPPC reaches OV at approximately 2.64 h.

Increasing charge rate exposes the limited baseline transport and voltage headroom: the reached fraction falls from 48.82% at 1C to 15.99% at 4C. Reached RMSE must be read together with coverage. A low error on an early prefix cannot establish agreement over the full experiment.

Initial 1C charge optimization

The initial optimization used the 1C charge record only. Four quantities were adjusted: initial graphite lithium fraction, initial LFP lithium fraction, added resistance and a graphite diffusivity multiplier. The initial fractions align the electrode operating windows, resistance affects current-dependent voltage loss, and diffusivity affects concentration polarization and relaxation. OCP shapes and the published exchange-current constants were held fixed.

The compact script loads the recorded CSV current directly and minimizes squared voltage residuals with SciPy least_squares. It fits 250 existing measurement rows and then computes the final error on all 6062 rows. Bounds prevent invalid initial fractions, negative diffusivity and resistance below 8 mΩ; early cutoff or solver failure receives a penalty. No new charge profile is synthesized.

Table 3 Initial 1C calibration and the compact example

Quantity	Starting guess	Archived 1C fit	Compact 1C fit
Initial graphite fraction	0.070000	0.075116	0.072941
Initial LFP fraction	0.800000	0.763522	0.763510
Added resistance mΩ	15.000	16.576	15.975
Graphite D multiplier	3.000	5.930161	4.535953
Graphite D m ² /s	9.000×10^{-15}	1.77905×10^{-14}	1.36079×10^{-14}

The archived original 1C study reported a reduction from 116.47 to 15.29 mV full-record RMSE. The packaged compact example achieved 15.15 mV full-record RMSE and 9.45 mV during its CC mask. Both use 1C charge alone; neither establishes transferability to discharge or HPPC. The compact optimizer starts from a 15 mΩ trial resistance, distinct from the 10 mΩ baseline assessment.

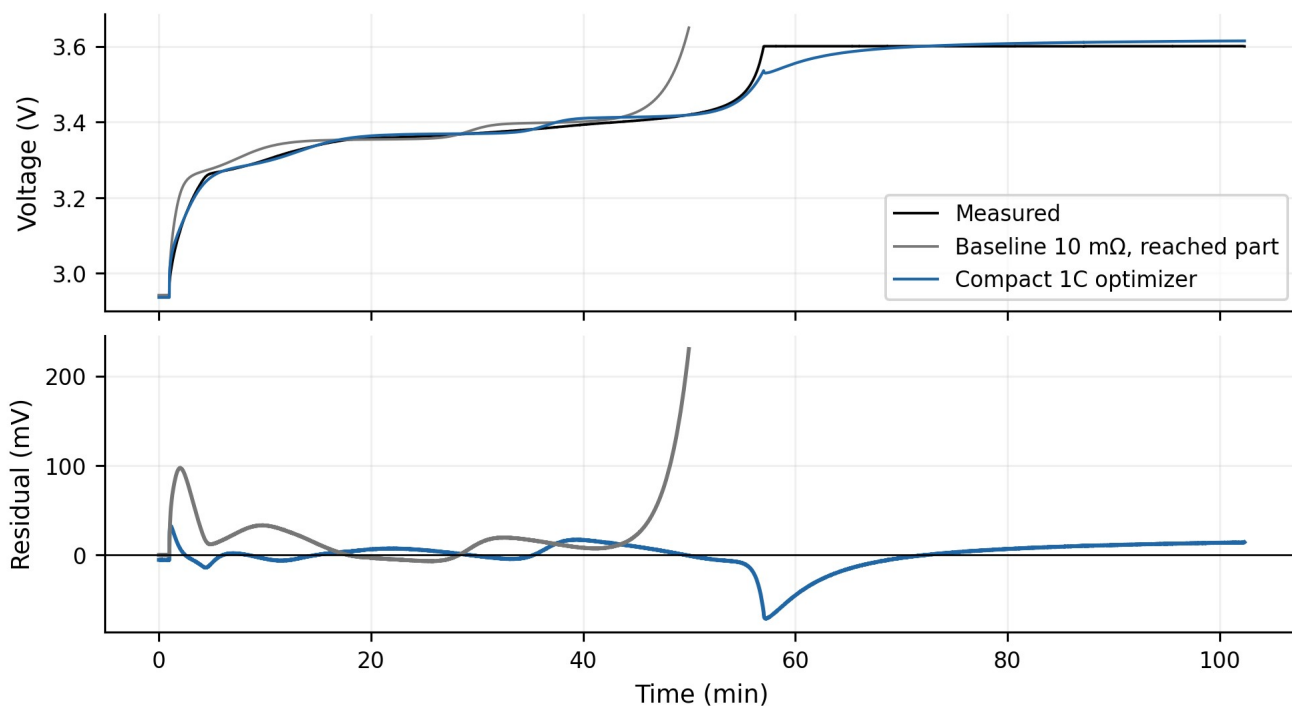


Figure 4 Compact 1C optimization voltage and residuals. The baseline line covers only its reached prefix; the fitted line covers the complete record. Residual is simulated minus measured voltage. The compact CC metric uses $||\cdot|| \geq 95\%$ of its peak.

For reproducibility, the subsequent grouped search retains the archived original 1C seed in `optimization_inputs/1C_optimized_parameters.json`. It does not silently replace that seed with the compact example's numerically different optimum. Before grouped tuning, the resistance was set to the previously selected 8 mΩ HPPC reference, while the archived diffusivity and lithium inventory were retained. This reference is explicitly listed alongside the final values below.

5 Multi profile optimization with pulse checks

A charge-only fit can use a parameter to compensate for an error in OCP, electrode balance or history. A larger effective resistance, for example, may help a charge trace while increasing voltage excursions during high-current charge and discharge pulses. Combining all profiles in a single undifferentiated RMSE can also let dense or long phases dominate shorter pulse-sensitive behaviour. We therefore used a staged search with explicit HPPC acceptance checks.

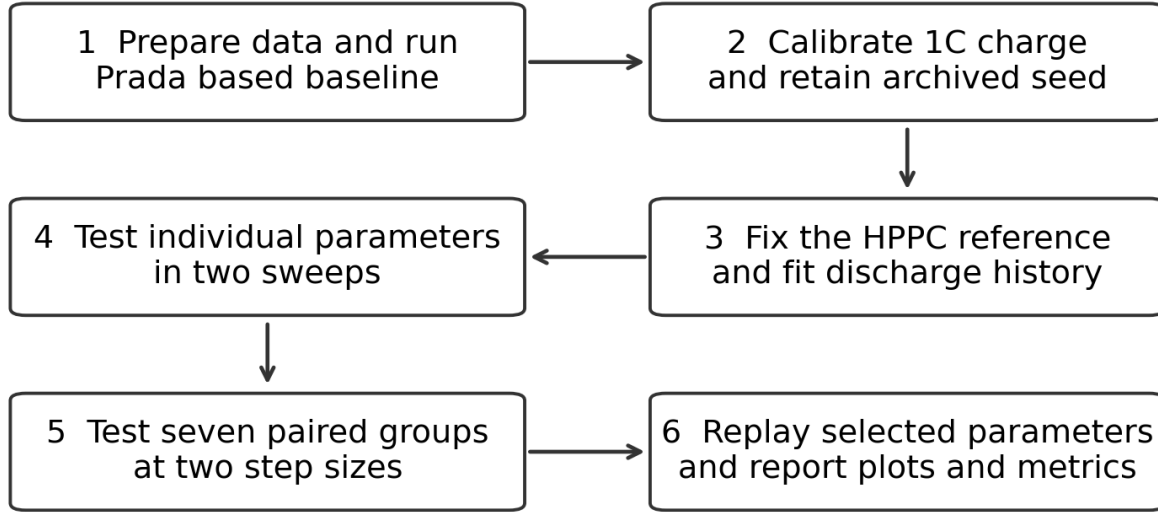


Figure 5 Sequence of the calibration study. The grouped search starts from the saved earlier 1C reference after the resistance selection. It estimates initial discharge history, then tests individual and paired parameter changes before detailed replay.

Local LFP history model

The final model uses PyBaMM's native one-state hysteresis submodel. A local history state h between -1 and $+1$ interpolates the LFP charge and discharge branches. The initial charge-branch fraction a_0 equals $(1 + h_0)/2$. The OCP relation and the implemented post-25.10 history convention are: [6,7]

$$U_p(y, h) = \frac{1+h}{2}U_{p, \text{ch}}(y) + \frac{1-h}{2}U_{p, \text{dis}}(y), \quad h_0 = 2a_0 - 1$$

$$\frac{dh}{dt} = \frac{\gamma|i_{\text{vol}}|}{2FC_s^{\text{max}}\epsilon_s} [\text{sgn}(i_{\text{vol}}) - h], \quad \gamma = \frac{2Q_p}{Q_h}, \quad Q_h = 0.05 \text{ Ah}$$

The local volumetric interfacial current density i_{vol} is the reaction current per electrode volume. For LFP, positive local reaction current corresponds to delithiation during charge and drives h towards $+1$; negative current during lithiation drives it towards -1 . The update speed depends on reaction passed locally. Where local reaction becomes negligible, the state retains its memory. Relaxation can reduce diffusion polarization without automatically removing this OCP history.

The fixed history charge scale Q_h is 0.05 Ah; it was a prior modelling choice informed by earlier HPPC assessment. Both LFP directions use the same decay parameter in this model. Charge and HPPC start at $h_0 = +1$. Discharge's fitted a_0 is 0.377065, giving $h_0 = -0.245869$. Graphite uses identical lithiation and delithiation OCP equations, so its hysteresis voltage amplitude is zero in the accepted model. The model adds effective LFP voltage memory; it does not resolve particle nucleation or phase fractions explicitly.

Parameters and fitting objective

Eight quantities were tested: added resistance R, graphite diffusivity Dn, graphite and LFP exchange-current densities, both electrode active fractions, lithium inventory and discharge initial history. R affects an immediate current-dependent contribution. Dn governs particle concentration gradients. Exchange-current densities affect reaction overpotential. Active fractions affect both electrode storage and reaction area. Lithium inventory adjusts the electrode operating balance, while initial history adjusts the discharge OCP branch mixture. These distinct effects motivated individual tests followed by related parameter pairs.

Particle radii, maximum concentrations, area, thickness, electrolyte properties and the history charge scale stayed fixed in this particular search. This limited the number of strongly coupled effective quantities fitted from full-cell voltage. Their retention does not establish that changing them would be useless; it defines the scope of the executed search.

Each of the five training profiles receives equal weight. Within charge profiles, CC, CV-like reduced-current operation and rest also receive equal weight; discharge uses active and rest phases. Up to 120 existing rows per phase are selected, while the complete current waveform is simulated. The charge CC mask is $|I| \geq 90\%$ of peak, and rest is $|I| \leq 0.05$ A. The remaining active charge rows form the CV mask. These are current-based labels, not a simulated voltage-control experiment.

$$J = 1000 \sqrt{\frac{1}{5} \sum_{p=1}^5 \left[\frac{1}{G_p} \sum_{g=1}^{G_p} \frac{1}{N_{p,g}} \sum_{i=1}^{N_{p,g}} e_{p,g,i}^2 + \{0.5(1 - c_p)\}^2 \right]}$$

In this expression, e is a voltage residual in volts, Gp is the number of available phases, Np,g is the selected sample count and cp is simulated end time divided by requested end time, limited to one. J is reported in mV-equivalent units. For an early cutoff, the objective holds the last predicted voltage for its sampled tail and adds the coverage term. Published voltage traces leave that tail missing. J is therefore a phase-balanced fitting score, not the full-record RMSE used in the results tables.

Search sequence and acceptance decisions

First, bounded scalar optimization estimates discharge's initial history fraction. Two coordinate sweeps then test downward and upward changes, with the second sweep using half-sized increments. Seven specified parameter pairs are finally tested at half and quarter of the original increments. Resistance is searched in mΩ, diffusivity and exchange-current densities in base-10 logarithms, and fractions and inventory in their physical units. Alternatives within one group start from the same group origin; the best passing alternative becomes the origin of the next group.

Table 4 Declared working parameter bounds

Quantity	Search bound
Added R	8-25 mΩ
Graphite D	10^{-15} - 10^{-12} m ² /s
Graphite / LFP j0	1-100 / 0.005-2 A/m ²
Graphite / LFP active fraction	0.48-0.62 / 0.28-0.52
Lithium inventory	2.2-3.4 Ah-equivalent
Discharge initial history a0	0-1

A candidate must improve the current best training objective by more than 0.05 mV and pass the first two hours of HPPC. A passing, improving candidate then undergoes full HPPC replay and a later-window check. Every HPPC comparison uses the fixed original reference, preventing gradual relaxation of the acceptance standard as tuning proceeds.

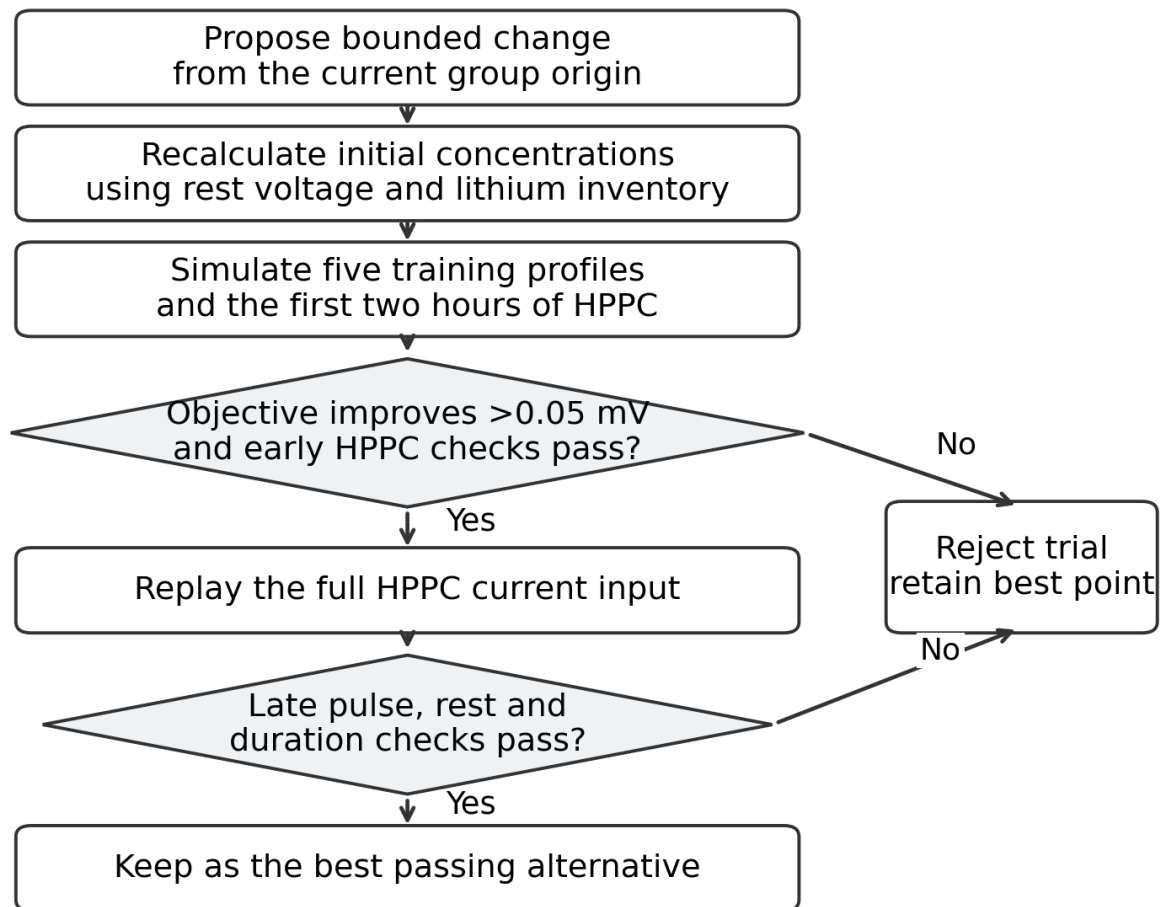


Figure 6 Decision procedure for each candidate. Solver or initialization errors reject a trial. After all alternatives in a group are assessed, its best passing point is carried forward; rejected trials do not replace that point.

Table 5 HPPC acceptance criteria

HPPC quantity	Maximum allowed deterioration
Charge and discharge pulse RMSE separately	Reference + 1 mV
Pulse peak absolute error separately	Reference + 5 mV
Rest RMSE	Reference + 2 mV
Stopping time	No more than 1 s earlier
Required predictions	Available throughout the comparison window

High-current samples use CSV current ≥ 5 A for charge and ≤ -5 A for discharge. Early means 0-2 h; late means 2 h to one second before the original reference cutoff. These allowances are declared engineering and numerical tolerances, rather than experimentally measured noise bounds. They permit small deteriorations within the limits and do not require every pulse to improve.

Table 6 Paired groups in their executed order

Pair	Joint trial directions	Outcome
1	Dn ↑ and LFP active fraction ↑	Half-step accepted at trial 32
2	Dn ↑ and graphite active fraction ↑	Half-step accepted at trial 34
3	Graphite j0 ↑ and LFP active fraction ↑	Half-step accepted at trial 36
4	Graphite j0 ↓ and LFP j0 ↑	Both alternatives rejected
5	LFP active fraction ↑ and inventory ↑	Both alternatives rejected
6	Graphite active fraction ↓ and inventory ↓	Quarter-step accepted at trial 43
7	Dn ↓ and graphite j0 ↑	Quarter-step accepted at trial 45

Half-step and quarter-step refer to fractions of the original search increment, not fractions of the parameter value. Pairing tests whether the combined change can pass where an isolated change may not. It does not assume that the two changes physically cancel each other.

Table 7 The seven accepted moves out of 45 proposals

Trial	Accepted change	Objective mV
1	Discharge initial history	40.047
25	Graphite active fraction increase	38.700
32	Dn and LFP active fraction increase	36.432
34	Dn and graphite active fraction increase	35.988
36	Graphite j0 and LFP active fraction increase	34.153
43	Graphite active fraction and inventory decrease	32.792
45	Dn decrease and graphite j0 increase	32.664

The initial objective was 251.562 mV before correcting discharge history; its early-cutoff penalty makes it much larger than an ordinary RMSE. A clear example of the pulse constraint is trial 44: its score of 32.552 mV was lower than trial 45's 32.664 mV, but trial 44 was rejected because the late discharge-pulse peak error exceeded the allowance. The smaller paired change in trial 45 passed. Resistance and LFP j0 remained unchanged after their alternatives were assessed.

6 Final results and physical states

Table 8 Baseline, grouped-search reference and accepted parameters

Quantity	Baseline	Search reference	Selected set
Added R mΩ	10.000	8.000	8.000
Graphite D m ² /s	3.000×10^{-15}	1.77905×10^{-14}	1.97565×10^{-14}
Graphite j ₀ A/m ²	7.500	7.500	8.32884
LFP j ₀ A/m ²	0.050	0.050	0.050
Graphite active fraction	0.5800	0.5800	0.5875
LFP active fraction	0.3740	0.3740	0.3940
Inventory Ah-equivalent	2.36705	2.73176	2.72676
Discharge history a ₀	1.0000	1.0000	0.377065

The archived 1C optimizer's resistance was 16.576 mΩ. The grouped-search reference already used the earlier pulse-study selection of 8 mΩ; its resistance reduction did not occur inside the 45-proposal search. Reference initial concentrations are recalculated from rest voltage and the archived inventory, rather than copied unchanged from the 1C charge start. The optimized active fractions are effective values; they were not directly measured from electrode samples.

Table 9 Final assessment using the accepted fixed parameter set

Profile	Final RMSE mV	Time coverage	Extent
Charge 1C	27.35	100.00%	Full record
Charge 2C	25.57	100.00%	Full record
Charge 3C	26.94	100.00%	Full record
Charge 4C	20.07	100.00%	Full record
Discharge 1C	26.49	100.00%	Full record
HPPC	35.82	72.53%	Reached prefix

All five charge/discharge records complete. The 1C charge RMSE is higher than the charge-only fit because this common set also supports discharge and pulse behaviour. The five complete-record errors are about 20-27 mV, but phase-specific errors can be larger; in particular, final 2C and 3C CV errors are approximately 45.60 and 47.16 mV.

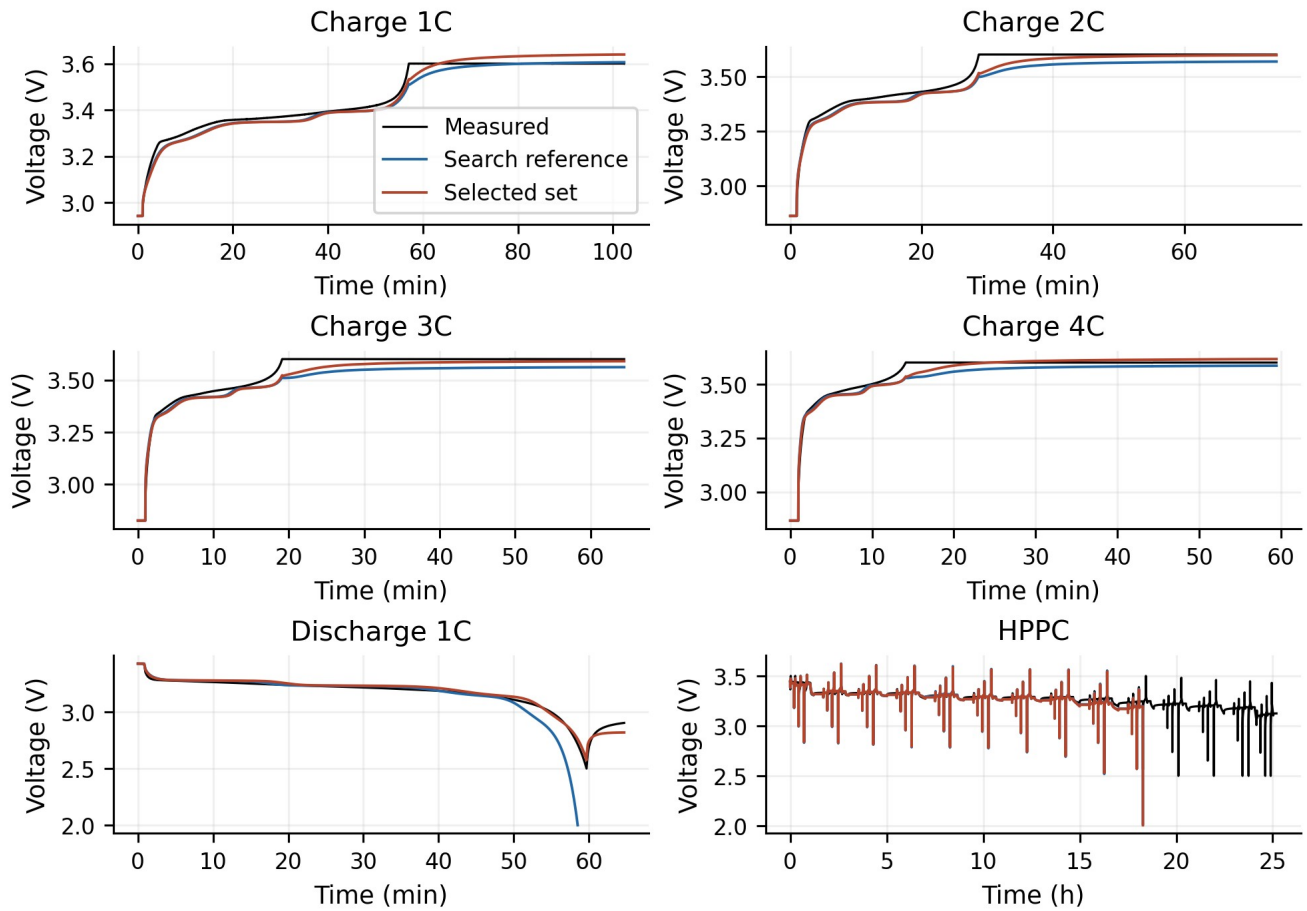


Figure 7 Measured voltage and predictions from the grouped-search reference and selected set. These detailed pipeline replays differ from the independent fixed-value assessment by at most 0.00765 mV in reported RMSE, owing to retained input and numerical details.

A fair baseline comparison uses identical measured times for both models, restricted to the baseline's reached window. Table 10 shows that the selected model reduces error on those common windows. Table 9 separately assesses its much longer simulated records; its errors should not be compared directly with baseline-prefix errors as if the durations were equal.

Table 10 Baseline and selected errors over identical baseline-reached windows

Profile	Baseline RMSE mV	Selected RMSE mV
Charge 1C	40.78	22.58
Charge 2C	41.03	23.62
Charge 3C	54.19	22.38
Charge 4C	64.10	20.74
Discharge 1C	219.63	18.12
HPPC	69.30	32.32

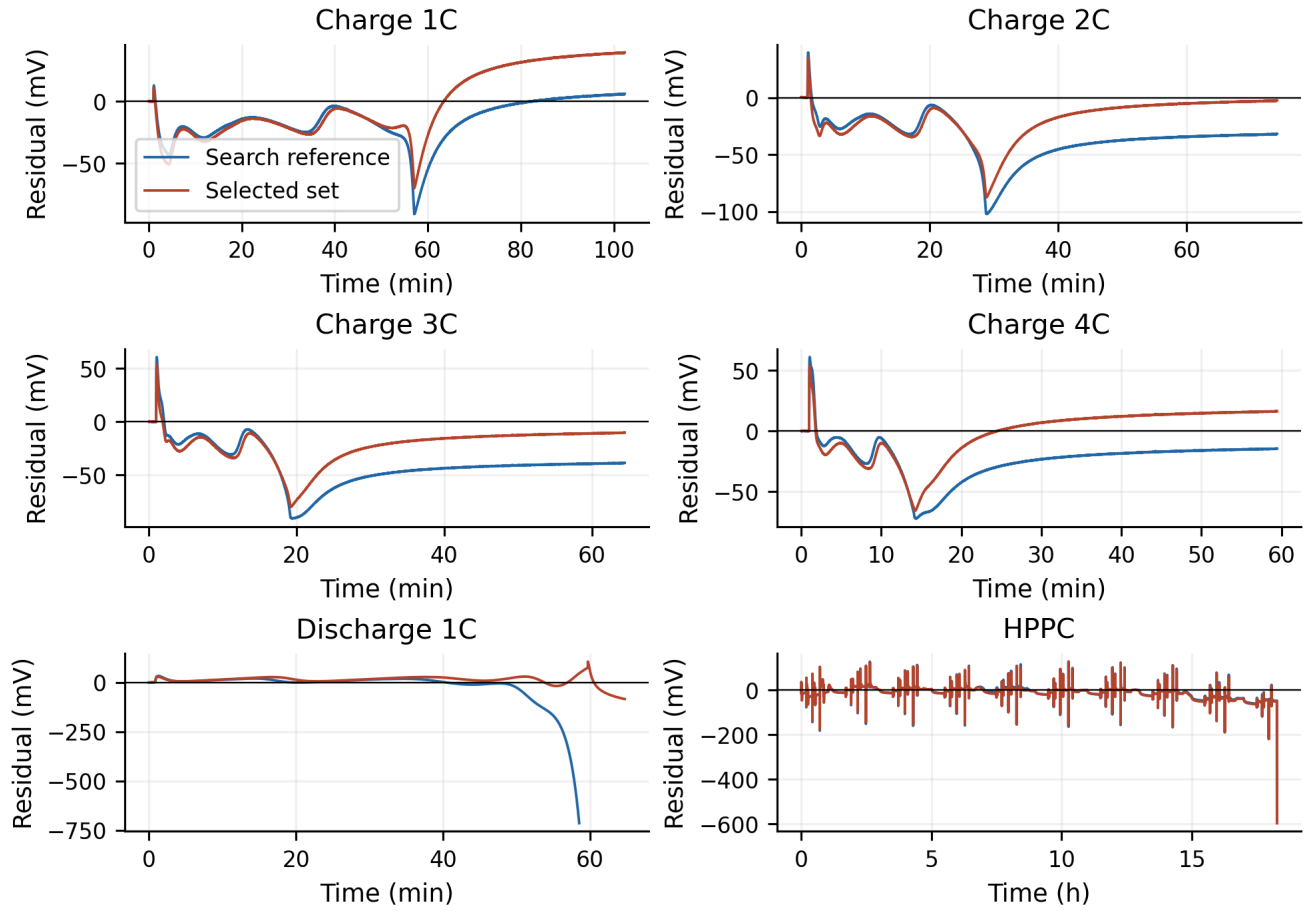


Figure 8 Voltage residuals for the same replays. Positive values indicate overprediction. The selected set improves several rate-dependent errors, while 1C charge and overall reached HPPC RMSE do not both improve relative to the search reference.

The residual plots also expose the tradeoff within the grouped search. The selected set corrects the premature 1C discharge cutoff and reduces the negative high-rate charge residuals, while its late 1C charge voltage becomes too high. The reached HPPC sample RMSE changes from 35.37 to 35.82 mV, despite improvement in the matched pulse groups. A phase-balanced objective with separate acceptance checks can therefore select a useful common set without improving every unweighted record metric.

HPPC assessment and choice of metrics

$$e_i = V_{\text{sim},i} - V_{\text{exp},i}, \quad \text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N e_i^2}$$

RMSE retains voltage units and penalizes larger residuals more strongly than MAE. It was reported together with MAE, bias, peak absolute error, phase or pulse errors and time coverage. No full-record HPPC RMSE is claimed because the simulation ends before the experimental record. Later predictions and residuals remain empty.

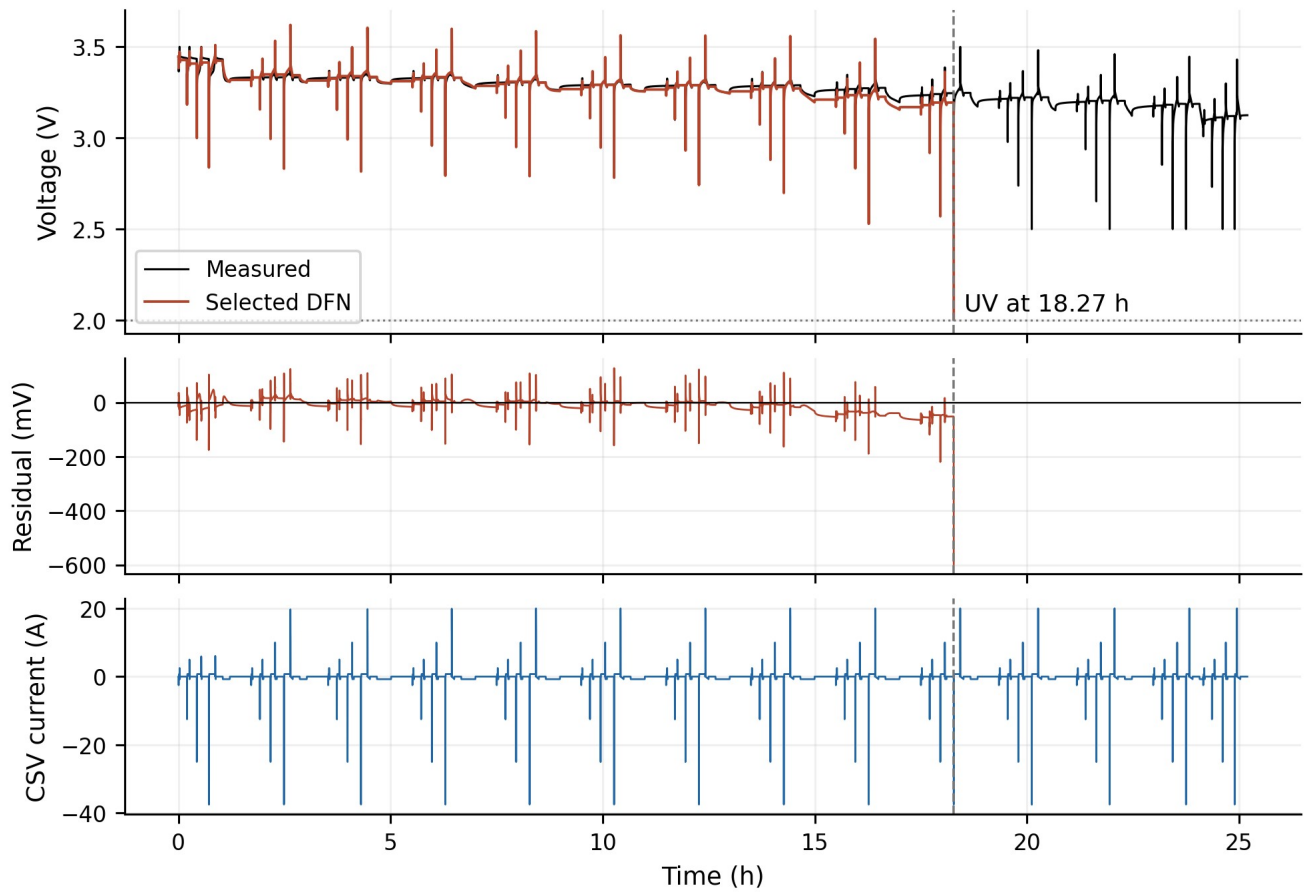


Figure 9 Full HPPC recording, reached model prediction, residuals and imposed current. The dashed line marks the simulated cutoff at 18.2736 h. Experimental measurements continue to 25.1950 h; no simulated voltage is drawn beyond the cutoff.

On the reached HPPC samples, RMSE is 35.82 mV, MAE is 24.80 mV and bias is -10.29 mV. The maximum absolute error is approximately 596.9 mV. Charge-pulse and discharge-pulse RMSE are 44.82 and 84.08 mV respectively. These values show why a single overall error can hide a substantial high-current mismatch. The reached time-weighted RMSE is 24.20 mV; its difference from sample RMSE reflects the uneven sampling of pulse and rest periods.

Table 11 Pulse errors on the fixed acceptance windows

Matched HPPC window	Reference pulse RMSE mV	Selected pulse RMSE mV
Early charge	24.26	23.80
Early discharge	63.06	58.24
Later charge	48.58	44.90
Later discharge	82.33	80.19

The early charge group contains only eight measured samples. Later-window and direction-specific sample counts are preserved in pulse_comparison.csv. HPPC guided earlier choices and the search acceptance procedure, so this is calibration assessment rather than independent holdout validation.

Anode potential assessment

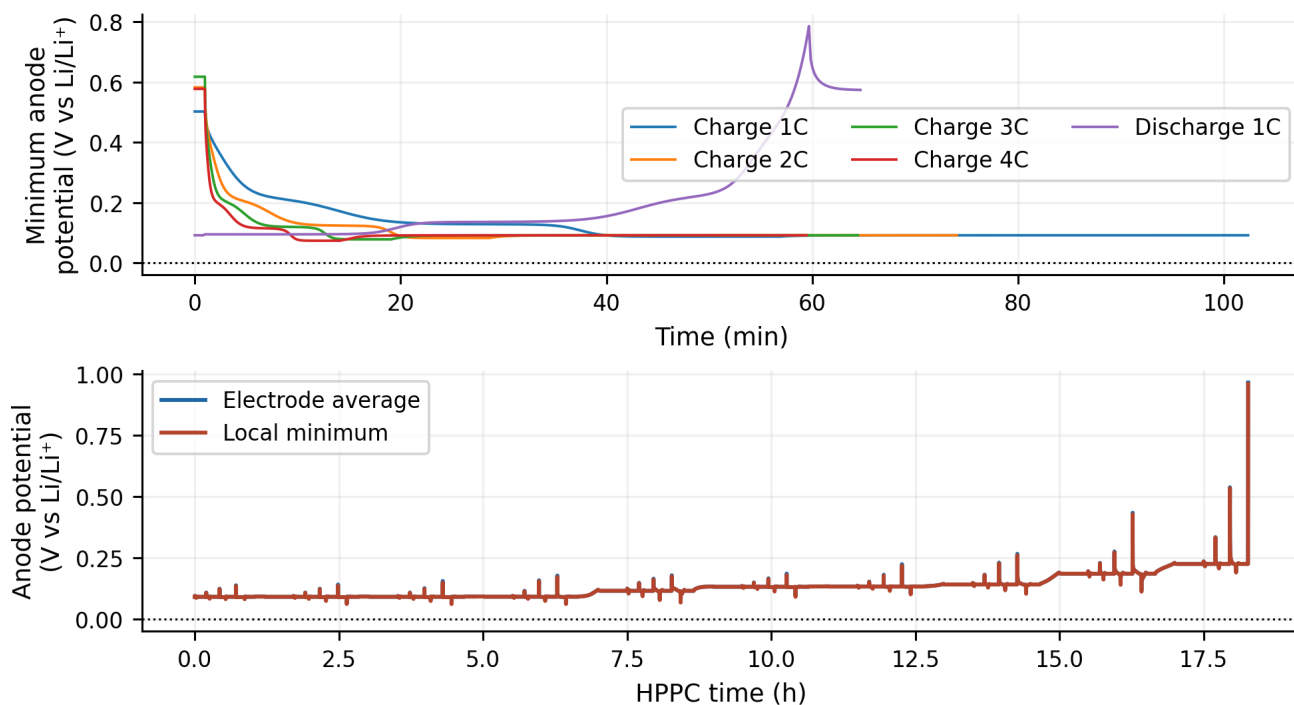


Figure 10 Anode surface potential difference versus Li/Li⁺. The upper panel shows local minima across the complete charge/discharge records. The lower panel shows the average and minimum anode potentials over the reached HPPC interval.

The minimum anode surface potential remains positive in these replays; the reached HPPC minimum is about 0.0624 V versus Li/Li⁺. This is a useful physical-state check, but it does not independently validate lithium-plating behaviour, since plating kinetics and target-cell electrode measurements were not included. The variable is the local solid-minus-electrolyte potential difference, and differs from graphite OCP by its reaction overpotential.

The simplified grouped search reproduces all 45 archived proposal labels, parameter vectors, decisions and rejection reasons, with seven accepted moves. Its selected parameters match the accepted fixed-value set to numerical precision. The reproduction record contains 222 passed checks, including initial voltage/lithium balance, parameter bounds, exported metrics and voltage events. These checks establish reproducibility of the implemented study, rather than uniquely identifying material properties.

7 Discussion of the remaining HPPC cutoff

The final model stops at 65784.868 s during a 37.5 A discharge pulse, about 15C on a nominal 2.5 Ah basis. The measured cell is still around 2.60 V when the model reaches 2.00 V. The stop is a configured minimum-voltage event. It demonstrates an excessive predicted voltage sag at that state, rather than agreement with the experimental stopping condition.

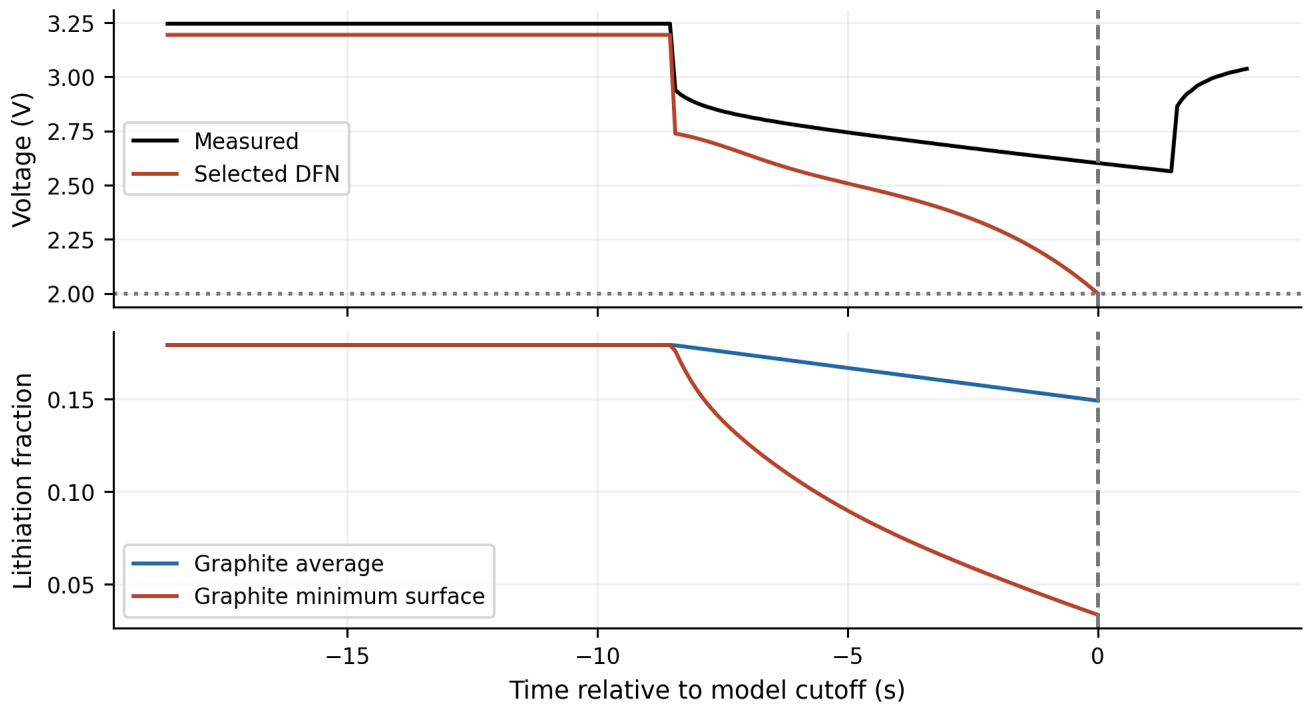


Figure 11 The final pulse leading to UV. Graphite particle surfaces become much less lithiated than the electrode average. Lithiation fractions are normalized electrode concentrations, not cell SOC. The measured voltage remains above the model cutoff.

Table 12 Voltage decomposition from the exported final HPPC states

Voltage contribution at the model event	Value V
Bulk OCV with the modelled history state	+3.140
Particle concentration contribution	−0.662
Added 8 mΩ resistance contribution	−0.300
Reaction contribution	−0.095
Other solid and electrolyte contributions	−0.083
Terminal voltage	2.000

The average graphite lithium fraction at this event is 0.149, while its particle-surface fractions are approximately 0.034. Lithium remains inside the particles, but the pulse removes it at the surface faster than the selected diffusion model replenishes it. The adopted graphite OCP gives about 0.258 V at the bulk fraction and 0.920 V at the depleted surface. Their difference accounts for about 0.661 V of the particle concentration contribution, which dominates this modelled event.

Because cell voltage contains the difference $U_p - U_n$, the rise in graphite potential at low surface lithiation sharply lowers terminal voltage. The characteristic diffusion time scales as particle radius squared divided by diffusivity. The selected radius, diffusivity and low-lithiation OCP shape therefore act together; the state trace cannot uniquely determine which should be changed to represent the actual cell. [2,8]

The same model completes the 1C discharge record with approximately 2.447 Ah of net discharge, whereas HPPC stops after approximately 2.087 Ah. Immediately before the failing pulse, model rest voltage is about 52 mV low; near cutoff, the voltage error grows to about 600 mV. These observations support a major pulse-polarization mismatch rather than attributing the event solely to total capacity. Initial inventory, electrode balance and assumed HPPC history can nevertheless contribute.

The voltage-matched HPPC start gives a graphite fraction of about 0.858, whereas the independent discharge start gives about 0.893 with its fitted history. This difference corresponds to roughly 0.103 Ah of graphite lithium in the model. It does not prove that HPPC initialization is wrong, but shows why matching starting voltage alone does not guarantee matching usable capacity across differently preconditioned records.

An 8 m Ω added resistance produces 0.300 V loss at 37.5 A, but it is not the largest loss at this event. Increasing fitted exchange-current density primarily addresses reaction overpotential, which is about 0.095 V here. These facts point first towards graphite concentration polarization and its OCP sensitivity, while retaining the prescribed resistance constraint.

Why the present search was stopped

The executed search was a finite local exploration with fixed parameter order, step sizes and seven paired groups. Its HPPC duration rule preserved the earlier 18.27-hour reference within one second; it did not explicitly require completing the 25.20-hour experiment. As a result, a candidate could pass the acceptance procedure without resolving the remaining tail. The completed search establishes a reproducible compromise, not a proof that further improvement is impossible.

Further fitting should explicitly prioritize the late low-SOC pulses and full-record coverage. Useful follow-up tests are graphite diffusivity or diffusion-time sensitivity, a bounded particle-radius or size-distribution study, and independent constraints on electrode capacity, initial history and OCP. All changes would still need to preserve the charge/discharge comparisons and the resistance floor. Additional parameter freedom should be supported by pulse relaxation or electrode measurements, because terminal-voltage fits alone can trade off transport, kinetics and electrode balance.

The main modelling limitations are the borrowed graphite OCP, constant effective kinetics and diffusivity, one representative particle size per electrode, uncertain initial microscopic history and approximate current reconstruction within logging gaps. Temperature variation was outside the supplied-data scope. Hysteresis improved the handling of voltage history, but the accepted scalar-state model does not establish a microscopic LFP phase-transition mechanism.

8 Conclusions and reproducibility

The Prada-based DFN baseline was improved through initial 1C calibration and a staged multi-profile search with explicit pulse checks. The accepted common set completes every supplied charge/discharge record with approximately 20-27 mV full-record RMSE. HPPC extends from the baseline's 2.64-hour OV event to an 18.27-hour UV event, with 72.53% time coverage and 35.82 mV reached-sample RMSE. Its late high-current discharge response remains the principal limitation.

The decision record explains how the accepted parameters were selected: improved training score, bounded parameters, voltage-consistent initial states, and separate early/late HPPC pulse and rest checks. The model is reproducible and useful for the assessed profiles; completion of the full HPPC record and independent validation remain future work.

Table 13 Programs supplied with the study

Program	Purpose
baseline_run.py	Baseline assessment of all six records
initial_parameter_optimization.py	Compact 1C charge optimizer
multi_profile_optimization_pipeline.py	Executed individual and paired search
optimization_model.py	Shared model and analysis functions
multi_profile_optimization.py	Replay of accepted charge/discharge parameters
hppc_and_all_profiles.py	Assessment of accepted parameters on all records

Extract the complete project folder and install the pinned requirements using Python 3.12. Before running the compact 1C example, set `CSV_FILE_PATH` to `data/Charge_1C.csv`, or another charge CSV path. The grouped pipeline must remain beside `optimization_model.py`, `data/` and `optimization_inputs/`. Replacing raw HPPC data also requires regenerating its processed waveform and charge audit. The README gives the run sequence; saved parameter JSON/CSV files, result traces and figures provide the reproducibility record.

The assignment also requires a GitHub repository link containing the source files, data references and documentation. The complete code folder, saved parameters, README and this report should be included in that submission.

References

- [1] Supplied assignment brief. Case Study LFP Cell Physical Modeling and Parametrization in PyBaMM. Three pages. Experimental inputs: `Charge_1C.csv`, `Charge_2C.csv`, `Charge_3C.csv`, `Charge_4C.csv`, `Discharge_1C.csv` and `HPPC_Converted.csv`.
- [2] PyBaMM documentation. Doyle-Fuller-Newman Model. Version 25.12.0. Governing equations and initial conditions. <https://docs.pybamm.org/en/v25.12.0/source/examples/notebooks/models/DFN.html>
- [3] Prada E, Di Domenico D, Creff Y, Bernard J, Sauvant-Moynot V and Huet F. A Simplified Electrochemical and Thermal Aging Model of LiFePO₄-Graphite Li-ion Batteries: Power and Capacity Fade Simulations. *Journal of The Electrochemical Society* 160, A616-A628, 2013. <https://doi.org/10.1149/2.053304jes>. Supplied paper; Table IV and the 2.3 Ah A123 cell description.
- [4] Afshar S, Morris K and Khajepour A. Efficient electrochemical model for lithium-ion cells. arXiv:1709.03970, 2017. <https://arxiv.org/abs/1709.03970>. Supplied paper; charge and discharge OCP equations on PDF page 6.
- [5] Chen CH, Brosa Planella F, O'Regan K, Gastol D, Widanage WD and Kendrick E. Development of Experimental Techniques for Parameterization of Multi-scale Lithium-ion Battery Models. *Journal of The Electrochemical Society* 167, 080534, 2020. <https://doi.org/10.1149/1945-7111/ab9050>. Supplied paper; LG M50 negative-electrode characterization.
- [6] PyBaMM 25.12.2 source distribution. `Prada2013.py`, `base_hysteresis_ocp.py` and `one_state_hysteresis_ocp.py`. The installed source establishes the Chen graphite / Kashkooli LFP kinetic provenance and the implemented history convention. Project source: <https://github.com/pybamm-team/PyBaMM>
- [7] PyBaMM documentation. Hysteresis State models. Version 25.12.0. <https://docs.pybamm.org/en/v25.12.0/source/examples/notebooks/models/hysteresis-state-models.html>. The one-state implementation cites Axén et al., Evaluation of hysteresis expressions in a lumped voltage prediction model of a NiMH battery system in stationary storage applications, *Journal of Energy Storage* 48, 103985, 2022; <https://doi.org/10.1016/j.est.2022.103985>. Used here as an effective history-state model, not as LFP-specific microscopic evidence.
- [8] PyBaMM documentation. Plot voltage components. Version 25.12.0. <https://docs.pybamm.org/en/v25.12.0/source/examples/notebooks/plotting/plot-voltage-components.html>
- [9] Saved study records. A123 Simple DFN Project results/ and `optimization_inputs/`, including parameter comparisons, voltage/state CSVs, `candidate_history.csv`, `guard_definition.json`, reconstruction audits and verification records. The separate controlled OCP screening results are held in `graphite_ocp_pre_1C`.